Physics 468/668: Statistical Physics **Spring 2019**

Lecture 23: Landau's theory

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23.1 A general expansion of free energy

As early as 1937 Landau attempted a unified description of all second-order phase transitions in the sense that the second derivatives of the free energy, namely the specific heat and the magnetic susceptibility (or isothermal compressibility, in the case of fluids), show a divergence while the first derivatives, namely the entropy and the magnetization (or density, in the case of fluids), are continuous at the critical point. He emphasized the importance of an order parameter m_0 (which would be zero on the high-temperature side of the transition and nonzero on the low-temperature side) and suggested that the basic features of the critical behavior of a given system may be determined by expanding its free energy in powers of m_0 (for we know that, in the close vicinity of the critical point, $m_0 \ll 1$). **He also argued that in the absence of the ordering field (h = 0) the up-down symmetry of the system would require that the proposed expansion contain only even powers of** m_0. Thus, the zero-field free energy $\psi_0 (= A_0/NkT)$ of the system may be written as

$$\psi_0(t, m_0) = q(t) + r(t)m_0^2 + s(t)m_0^4 + \cdots \quad \left(t = \frac{T - T_c}{T_c}, |t| \ll 1\right) \tag{23.1}$$

The equilibrium value of the order parameter is then determined by minimizing ψ_0 with respect to m_0; retaining terms only up to the order displayed in eq. 23.1, which for thermodynamic stability requires that $s(t) > 0$ (if we want to have ψ reach minimum), we obtain

$$r(t)m_0 + 2s(t)m_0^3 = 0 \tag{23.2}$$

The equilibrium value of m_0 is thus either 0 or $\pm\sqrt{[-r(t)/2s(t)]}$. The first solution is of lesser interest, though this is the only one we will have for $t > 0$; it is the other solutions that lead to the possibility of spontaneous magnetization in the system. To ensure that m_0 has nonzero solutions, it needs that $r(t) <0$. When $r(t) <0$, $m_0 = 0$ is no longer a stable solution, whereas it will move to other adjacent solutions.

$$|m_0| = \sqrt{-(r(t)/2s(t)} \quad (t \leq 0) \tag{23.3}$$

We see that, for $t \geq 0$, there is only one minimum, which is located at $m_0 = 0$; for $t = 0$, the minimum is rather flat. For $t < 0$, on the other hand, we have two minima, located at $m_0 = \pm m_s$, as given by expression, with a maximum at $m_0 = 0$. Now, since ψ_0 has to be convex in m_0, so that the susceptibility of the system be nonnegative. We must replace the nonconvex portion of the curve, which lies between the points $m_0 = -m_s$ and $m_0 = m_s$, by a straight line. The replacement is reminiscent of the *Maxwell construction*.

Now we can also consider the expression of $r(t)$ and $s(t)$ near T_c. Ideally, they are both very small terms which can be considered as Taylor expansion series,

$$r(t) = \sum_{k\geq 0} r_k t^k, \quad s(t) = \sum_{k\geq 0} s_k t^t, \cdots \tag{23.4}$$

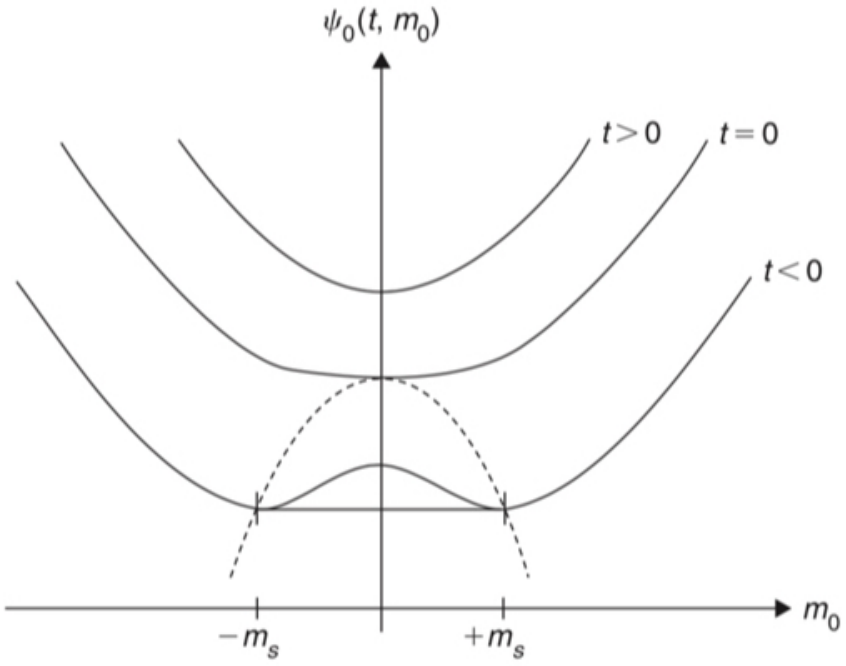


Figure 23.1: The free energy $\psi_0(t, m_0)$ of the Landau theory, shown as a function of m_0, for three different values of t. The dashed curve depicts spontaneous magnetization $m_s(t)$, while the horizontal line for $t < 0$ provides the Maxwell construction.

As you can see from Figure 23.1, at $t = 0$, it must satisfy that $r(t) = 0$ to ensure that there would be only one free energy minimum at $m = 0$. Therefore, a expansion can be

$$r(t) = r_1 t, \quad s(t) = s_0 \tag{23.5}$$

Plug it to Eq. 23.3, we get

$$|m_0| = \sqrt{-(r_1/2s_0)}|t|^{1/2} \quad (t \leq 0) \tag{23.6}$$

This gives that $\beta = 1/2$.

If $r(t)$ is a smooth function then the equilibrium value of m changes continuously from m=0 when $r(t) > 0$ to $m \neq 0$ at $r(t) < 0$. This is called 2nd order phase transition occurring at T_c with $r(T_c) = 0$.

Once we know the equilibrium value of m, we can substitute it back to the Taylor expansion

$$\psi_0(t, m_0) = \begin{cases} q(t) & t > 0 \\ q(t) - r(t)^2/2s(t) + r(t)^2/4s(t) = q(t) - r(t)^2/4s(t) = q(t) - r(t)^2/4s_0 & t < 0 \end{cases} \tag{23.7}$$

Because $r(0) = 0$, A is continuous at T_c, $S = -\partial A/\partial T$ is also continuous. But if you differentiate A twice, the term $r(t)'/2s_0$ is generally not vanishing. It means that the 2nd order derivative of A, such as C_V changes discontinuously.

23.2 Under an ordering positive field h

Now we consider there exists an ordering positive field h. If the field is weak, the only change in the expression for the free energy would be the addition of a term $-hm$.

$$\psi_h(t,m) = -hm + q(t) + r(t)m^2 + s(t)m^4 \tag{23.8}$$

The equilibrium value of m is

$$\frac{\partial \psi}{\partial m} = -h + r(t)m + 4s(t)m^3 = 0 \tag{23.9}$$

The low-field susceptibility of the system, in units of $N\mu^2/kT$, turns out to be

$$\chi = \left(\frac{\partial h}{\partial m}\right)_t^{-1} = \frac{1}{2r(t) + 12s(t)m^2} \tag{23.10}$$

valid in the limit $h \to 0$. Now for $t < 0$, $m \to 0$, we get

$$\chi \approx 1/2r_1 t \quad (t \geq 0) \tag{23.11}$$

so the critical component (γ) is 1.

For $t < 0$, $m \to \sqrt{[(r_1/2s_0)|t|]}$, we get

$$\chi \approx 1/4r_1|t| \quad (t \leq 0) \tag{23.12}$$

So γ becomes 1.

Finally, we set $t = 0$, leading to

$$h \approx 4s_0 m^3 \quad (h \to 0) \tag{23.13}$$

giving $\delta = 3$.

23.3 Heat capacity

From the thermodynamic integral, we have $dA = -SdT + PdV$ (note that PdV term in the Ising model can be considered as HdM)

$$\frac{\partial A}{\partial T} = -S, \quad \frac{\partial^2 A}{\partial T^2} = -\frac{\partial S}{\partial T} = -\frac{C}{T}, \tag{23.14}$$

In the first equation, we have $A_0/Nk = \psi_0 T$,

$$\frac{1}{Nk}\frac{\partial^2 A}{\partial T^2} = \frac{\partial^2 \psi}{\partial T^2} + 2\frac{\partial \psi}{\partial T} \tag{23.15}$$

Therefore,

$$C/T \approx -Nk\left(T_c \frac{\partial^2 \psi}{\partial t^2} + 2\frac{\partial \psi}{\partial t}\right) \tag{23.16}$$

So,

$$C \approx -NkT_c^2 \frac{\partial^2 \psi}{\partial t^2} = -NkT_c^2(2q_2 + 6q_3 t + \cdots) \approx -2q_2 NkT_c^2 \tag{23.17}$$

It means that $\alpha = 0$.

The most striking feature of the Landau theory is that it gives exactly the same critical exponents as the mean field theory or the van der Waals theory. Actually it goes much further, for it starts with an expression for the free energy of the system containing parameters $q_k, r_k, s_k, \cdots$, which represent the structure of the given system and the interactions operating in it, and goes on to show that, while the amplitudes of the various physical quantities near the critical point do depend on these parameters, the critical exponents do not!